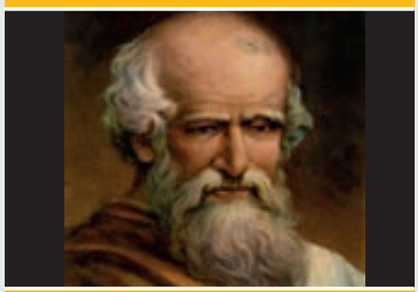
3 Integral Calculus – II



Archimedes
(287 BC(BCE)-217 BC(BCE))

Introduction

History of integration begins over 2500 years ago. The first Greek Mathematician Antiphon (around 430 BC (BCE)). introduced the "method of exhaustion" to find areas of simple polygons and more complicated curves. (method of exhaustion means dividing the given area into infinite number of triangles). Though Antiphon invented the method of exhaustion to find area bounded by complicated curves, the mathematician Eudoxus did the logical development in this method of exhaustion. Later Euclid used this method to calculate the area of circle.

Using the same method of exhaustion Archimedes (287 BC(BCE)-217 BC(BCE)) find the area bounded by parabola. Thus using integration area bounded by curves is developed. We will see the method of finding area by using integration in this chapter.

Learning Objectives

After studying this chapter , the students will be able to understand

- the geometrical interpretation of definite integral.
- the applications of integration in finding the area bounded by a curve.
- the concept of consumer's & producer's surplus.
- the applications of integration in Economics and Commerce.

3.1 The area of the region bounded by the curves

Using integration we can evaluate the area bounded by the curves with coordinate axes. We can also calculate the area between two given curves.

3.1.1 Geometrical Interpretation of Definite Integral as Area under a curve:

Suppose we want to find out the area of the region which is bounded above by a curve $y = f(x)$, below by the $x-axis$ and the lines $x = a$ and $x = b$.

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Now from Fig 3.1 let the interval $[a\ b]$ is divided into n subintervals $[x_{i-1},\ x_i]$ of equal length Δx_i *i.e* $x_i - x_{i-1} = \Delta_{xi}$ for any $x_i' \in [x_{i-1},\ x_i]$ let $f(x_i')$ be the height of

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